Low-pass and High-pass Filtering of Sentinel-2 Imagery over the Aravalli Ridge

Which filtering technique suits which remote sensing application?

Group 1

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EES403: Advanced Remote Sensing and GIS
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Introduction

Presented by **Robin**

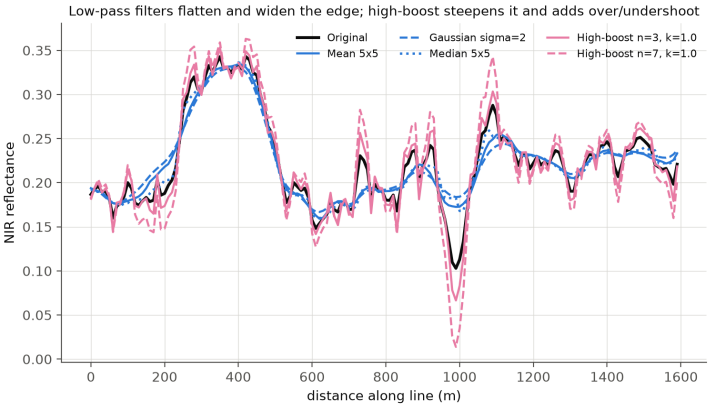
Low-pass smooths, high-pass sharpens: which one, for what?

Spatial frequency: how often brightness changes per unit distance (Lillesand et al., 2015).

Low-pass keeps slow change fields, water ⇒ smoother image, less noise

High-pass keeps fast change roads, ridge crests ⇒ edges stand out

We test both on five jobs: noise removal, land-cover classification, edges and lineaments, change to the spectra, speed.

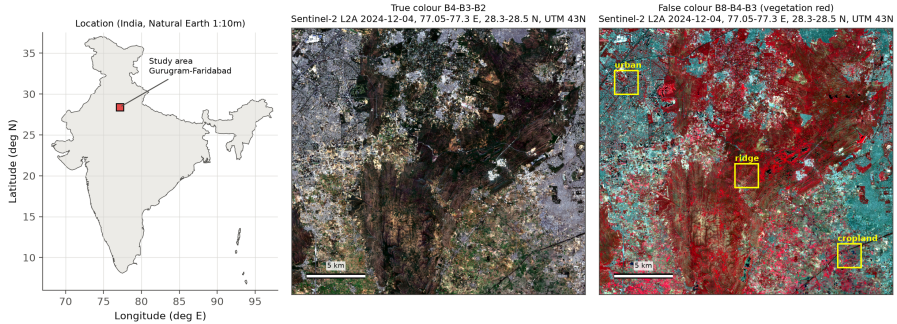


NIR reflectance along a 1.6 km line on the ridge: original (black), low-pass (blue), high-boost (pink).

Study Area and Data

Presented by **Sangam**

One winter scene holds ridges, city and farmland



Left: location. Centre: true colour. Right: false colour, vegetation red. Yellow boxes: ridge, urban and cropland windows.

Image	Sentinel-2 L2A, 2024-12-04, tile 43RGM	Area	25 × 23 km, 2491 × 2261 px
Bands	blue, green, red, near-infrared (10 m)	Cloud and haze	0.08% (best of 11 scenes)
Reference map	ESA WorldCover 2021, 5 classes	Water	0.19% of area, dropped

Methods

Presented by **Pranay**

Spatial domain: each output pixel is a weighted sum of its neighbours

$$g(x,y) \;=\; \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)\, f(x-s,\; y-t)$$

computed by shift and accumulate over the kernel cells, with mirror padding at the border

Low-pass

Mean $n \times n$: $w = 1/n^2$, $n = 3, 5, 7, 9$

Gaussian: $w(s,t) \propto \exp\!\big(-\frac{s^2+t^2}{2\sigma^2}\big)$, $\sum w = 1$, $\sigma = 1, 2, 3$

Median 3×3, 5×5: non-linear, ignores single spikes

High-pass

$\mathrm{HPF}_n = f - \mathrm{mean}_{n\times n}(f)$ (Lillesand et al., 2015)

Laplacian $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$; Sobel, Prewitt: $\sqrt{G_x^2 + G_y^2}$

High-boost: $f + k \cdot \mathrm{HPF}_n$, $k = 0.5, 1, 1.5$

Written in NumPy; unit tests against `scipy.ndimage` agree to $< 10^{-5}$. Gonzalez and Woods (2018), Ch. 3.

Frequency domain: filtering is a multiplication of the spectrum

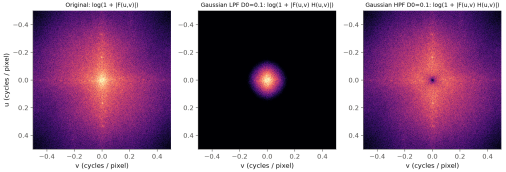
$$g = \mathcal{F}^{-1}\big\{H(u,v)\cdot\mathcal{F}\{f\}\big\}$$

D: distance from spectrum centre, $D = 1$ at Nyquist; $D_0 = 0.1, 0.2, 0.3$

Ideal:	$H = 1$ if $D \le D_0$, else 0
Butterworth $(n = 2)$:	$H = 1/\big(1 + (D/D_0)^4\big)$
Gaussian:	$H = \exp\big(-D^2/2D_0^2\big)$
High-pass:	$H_{\mathrm{HP}} = 1 - H$

Gaussian mask ≡ spatial Gaussian with $\sigma = 1/(\pi D_0)$ px

Ideal mask: sharp cut ⇒ sinc kernel ⇒ ringing (≈ 9% overshoot)



Log-magnitude spectra of the NIR crop: original, Gaussian LPF, Gaussian HPF ($D_0 = 0.1$).

In total 43 filter settings, plus the course example (square ideal masks of 61, 91, 121 px).

How each experiment is scored

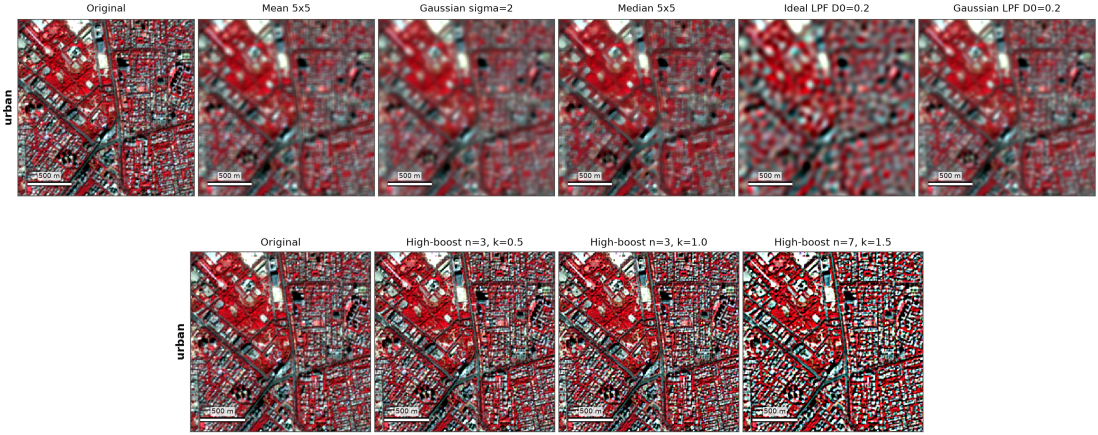
	Question	Deciding measure
E1	Noise removal	PSNR $= 10\log_{10}(1/\mathrm{MSE})$ dB on a noisy crop; also SSIM (Wang et al., 2004)
E2	Land-cover classification	Random Forest (Breiman, 2001); Cohen's $\kappa = (p_o - p_e)/(1 - p_e)$; salt-and-pepper index
E3	Edges and lineaments	Canny (Canny, 1986) edge F1 vs WorldCover boundaries; ridge coherence $C = \dfrac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}$
E4	Change to the spectra	spectral angle $\mathrm{SAM} = \arccos \dfrac{\mathbf{x}\cdot\mathbf{y}}{\lvert\mathbf{x}\rvert\lvert\mathbf{y}\rvert}$; $\lvert\Delta\mathrm{NDVI}\rvert$
E5	Speed	median of 5 runs on one full band, spatial vs FFT

E2 uses a **spatially blocked split** (128 px blocks, 70/30) so test pixels are not near-copies of training pixels.

Results I: What the Filters Do

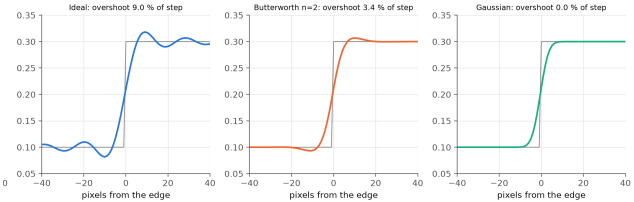
Presented by **Aparna**

Low-pass blurs the street grid; high-boost makes it crisper



Gurugram window, false colour, same stretch. Top: low-pass filters. Bottom: high-boost (original plus a share of the edge layer).

The ideal mask rings; the median is best for impulse noise



Clean step edge passed through each mask, $D_0 = 0.1$.

Mask	Ideal	Butterworth	Gaussian
Overshoot (% of step)	**9.0**	3.4	0.0

E1: noise removal (PSNR, higher is better)

Salt-and-pepper noise
noisy 21.01 dB → median 3×3 **36.49 dB**
best averaging filter: 31.06 dB

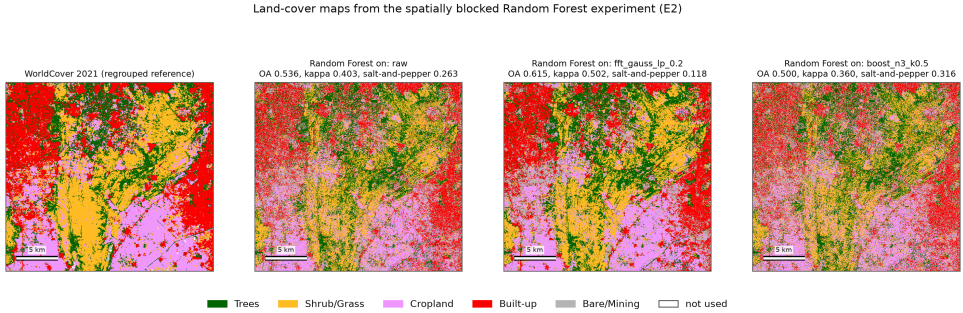
Random noise
noisy 33.98 dB → Gaussian $\sigma = 1$ **35.56 dB**
13 of 18 low-pass filters made it worse

Use smooth masks, not the ideal one. Use the median for spikes, and only light smoothing for random noise.

Results II: Classification, Edges and Cost

Presented by **Sangeeta**

Smoothing helps classification; sharpening helps edges



E2 classification (κ, higher is better)			**E3 edges** (boundary F1, higher is better)	
No filter	0.403	speckle 0.263	No filter	0.675
Best low-pass	**0.502**	speckle **0.118**	Best high-boost	**0.742**
Best high-boost	0.360		Mean 5×5	0.247

Cost: light filters change the spectral angle by 1.20–2.30°; a full band takes 0.10 s with a 3×3 kernel and 1.36–1.48 s with any FFT mask.

Conclusions

Presented by **Robin**

Decision: choose the filter by the job

Application	Best filter	Evidence (best vs. no filter)
Noise removal, random noise	Gaussian $\sigma = 1$	PSNR (dB): **35.56** vs 33.98
Noise removal, impulse noise	Median 3×3	PSNR (dB): **36.49** vs 21.01
Land-cover classification	Gaussian LPF $D_0 = 0.2$	Cohen's κ: **0.502** vs 0.403
Lineament mapping (ridges)	Prewitt	ridge coherence: **0.536** vs 0.510
Urban boundary delineation	High-boost $n = 3$, $k = 0.5$	built-up edge F1: **0.545** vs 0.543
Visual interpretation	High-boost $n = 3$, $k = 1.0$	NIR mean gradient: **0.0417** vs 0.0245

Pixel values feed a calculation

light **Gaussian**; **median** 3×3 for spikes

Edges and structure are the point

high-boost $k = 0.5$–1; **Sobel/Prewitt** layer

Avoid the ideal mask (ringing) and heavy smoothing. Limits: one scene, one area; WorldCover is a model output; gains of about 0.01 are not a ranking.

Thank you. Questions?

Live demo: `python -m rsfilter demo` (offline, 512×512 sample)

References I

Josef Bigün and Gösta H. Granlund. Optimal orientation detection of linear symmetry. In *Proceedings of the First International Conference on Computer Vision*, pages 433–438, London, 1987.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

John Canny. A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 8(6): 679–698, 1986.

M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, A. Meygret, F. Spoto, O. Sy, F. Marchese, and P. Bargellini. Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120:25–36, 2012.

European Space Agency. Sentinel-2 Level-2A Product Specification and Sen2Cor Algorithm Theoretical Basis Document. ESA Sentinel Online.

Rafael C. Gonzalez and Richard E. Woods. *Digital Image Processing*. Pearson, New York, 4th edition, 2018.

John R. Jensen. *Introductory Digital Image Processing: A Remote Sensing Perspective*. Pearson, Glenview, IL, 4th edition, 2015.

Thomas M. Lillesand, Ralph W. Kiefer, and Jonathan W. Chipman. *Remote Sensing and Image Interpretation*. Wiley, Hoboken, NJ, 7th edition, 2015.

Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

Daniele Zanaga, Ruben Van De Kerchove, Dirk Daems, Wanda De Keersmaecker, Carsten Brockmann, Grit Kirches, Jan Wevers, Oliver Cartus, Maurizio Santoro, Steffen Fritz, et al. ESA WorldCover 10 m 2021 v200. Zenodo, 2022.